Norme:

→ vectoriale: $\|x\|_p = \sqrt[p]{\sum_{i=1}^{n} |x_i|^p}$, $p \geq 1$

↳ norm(x, p)

→ norma infinit: $\|x\|_\infty = \max_{i=1:n} \{|x_i|\}$

norma sumă: $p=1$: $|x_1|+|x_2|+\dots$

↳ euclidiană: $p=2$: $\sqrt{x_1^2+x_2^2+\dots}$

→ matriceale:

→ Frobenius: $\|A\|_F = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2}$ → norm(A, 'fro')

→ $\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|}$ (norma vect)

2) $\|A\|_1 = \max_{j=1:n} \{\sum_{i=1}^{m} |a_{ij}|\}$

$\|A\|_\infty = \max_{i=1:m} \{\sum_{j=1}^{n} |a_{ij}|\}$

$\|A\|_2 = \sqrt{\rho(A^T \cdot A)}$

↓ maximul valorilor proprii în modul

Tipuri de matrice:

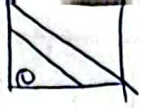
→ diagonală: diag(A, k)
diag(x, k)

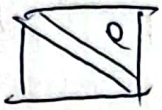
→ tridiagonală: diag(d,-1)+diag(a)+diag(c,1)

→ Δ → inf: triu(A, k), $k \geq 0$

↳ sup: triu(A, k), $k \leq 0$

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→ Hessenberg → superior :  → trim(A, -1)

→ inferior :  → trim(A, 1)

(strict)

→ diagonal dominantă pe linii (pe coloane e similar)

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|, \forall i = 1:n$$

$$\text{col}: |a_{jj}| \geq \sum_{\substack{i=1 \\ i \neq j}}^{n} |a_{ij}|, \forall j = 1:n$$

→ pozitiv definită :

→ mat. pătratică

→ mat. simetrică ($A = A^T$)

→ $\boxed{x^T \cdot A \cdot x > 0}$, $\forall x \in \mathbb{R}^n$

→ verificare dacă mat. A e poz. def :

ieși un vect. x oarecare $[x_1\ x_2 \dots x_n]^T$

SAU ți faci efectiv calculele $x^T \cdot A \cdot x$

→ Sylvester : mat. poz. def. ($\Leftrightarrow$)

($\Leftrightarrow$) A pătratică, simetrică și

TOȚI MINORII PRINCIPALI sunt > 0

→ ortogonale : ⇝ vezi pag. următoare

Matrice ortogonală

$Q^{-1} = Q^T$

Proprietăți: 1) $\det Q = \pm 1$

2) $\|Qx\|_2 = \|x\|_2$

3) $\|Q\|_2 = 1$

4) $\text{cond}\, Q = \|Q\|_2 \cdot \|Q^{-1}\|_2 = 1$

5) $\|QA\|_2 = \|A\|_2$

Matrice Hp (Householder) – proprietăți:

1) Hp simetrică

2) Hp ortogonală

Matrice G_{ke} (Givens) – proprietăți:

1) G_{ke} ortogonală

2) G_{ke} matrice de rotație (cu θ k de rotație; $c = \cos\theta$, $s = \sin\theta$)

3) G_{ke} diferă de Im prin 4 elemente

$G_{ke}(k,k) = c$

$G_{ke}(k,l) = -s$

$G_{ke}(l,k) = s$

$G_{ke}(l,l) = c$

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Metode iterative de rezolvare a SEL

$Ax = b$

$A = N - P$

$(N-P)x = b \Rightarrow x = N^{-1}Px + N^{-1}b$

$x^{(p+1)} = \underbrace{N^{-1}P}_{G} x^{(p)} + \underbrace{N^{-1}b}_{c}$

$G = N^{-1}P \Leftarrow x^{(p+1)} = Gx^{(p)} + c$

$e^{(p)} = x^{(p)} - x^*$ → eroarea aproximării sol. la pasp

→ * metodă iterativă converge ⇔ $\rho(G) < 1$

→ A diagonal dominantă pe linii ⇒ met. converge

$A = D - L - U$

toleranța: $\varepsilon = \left| \frac{x^{(p+1)} - x^{(p)}}{x^{(p+1)}} \right|$

Jacobi: $N = D$, $P = L + U$ $\Rightarrow G_J = D^{-1}(L+U)$

Gauss-Seidel: $N = D - L$, $P = U$ $\Rightarrow G_{GS} = (D-L)^{-1}U$

Ce aleg?

mat. oarecare ⇒ suprarelaxare

superior Hessenberg ⇒ suprarelaxare / Gauss-Seidel

Σ diag(vect, m) ⇒ ——— m ———

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Conditionarea matricelor:

$$\text{cond}(A) = \|A\|_\infty \cdot \|A^{-1}\|_\infty$$

$\text{cond}(A) \approx 1 \Rightarrow$ well-conditioned

$\text{cond}(A) >> 1 \Rightarrow$ ill-conditioned

Alg:

stabili:

- Cholesky
- Gram-Schmidt modificat
- Householder
- Thomas?

instabili:

- Crout
- Doolittle
- Gram-Schmidt
- eliminare gaussiană (fără pivotare)

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Gram-Schmidt clasic vs. modificat :

- pe coloane : $j = 1:n$
  - calcul r_{kj}, $k = 1:j-1$
  - $aux = q_j - \sum_{k=1}^{j-1} r_{kj} \cdot q_k$
  - $r_{jj} = norm(aux)$
  - $q_j = \frac{aux}{r_{jj}}$;

- pe linii : $i = 1:k$
  - $r_{ii} = norm(a_i)$
  - $q_i = \frac{a_i}{r_{ii}}$
  - Nr. ac. fp
    - calcul r_{ij}
      $a_j = a_j - q_k \cdot r_{kj}$

SST - deducere formulă

$$Ux = b$$

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots & u_{1n} \\ & u_{22} & u_{23} & \cdots & u_{2n} \\ & & \ddots & & \vdots \\ 0 & & & \ddots & \vdots \\ & & & & u_{nn} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ \vdots \\ b_n \end{bmatrix}$$

$$\begin{cases} u_{11}x_1 + u_{12}x_2 + u_{13}x_3 + \cdots + u_{1n}x_n = b_1 \\ u_{22}x_2 + u_{23}x_3 + \cdots + u_{2n}x_n = b_2 \\ \cdots\cdots\cdots\cdots\cdots \\ u_{nn}x_n = b_n \end{cases}$$

$$\rightarrow \begin{cases} x_n = \dfrac{b_n}{u_{nn}} \\ x_{n-1} = \dfrac{b_{n-1} - u_{n-1,n}x_n}{u_{n-1,n-1}} \\ \vdots \\ x_2 = \dfrac{b_2 - u_{23}x_3 - u_{24}x_4 - \cdots - u_{2n}x_n}{u_{22}} \\ x_1 = \dfrac{b_1 - u_{12}x_2 - u_{13}x_3 - \cdots - u_{1n}x_n}{u_{11}} \end{cases}$$

$$i = n:-1:n; \quad x_i = \frac{b_i - \sum_{k=i+1}^{n} u_{ik} \cdot x_k}{u_{ii}}$$

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SIT - deducere formulă

$$Lx = b$$

$$\begin{bmatrix} l_{11} & & & & \\ l_{21} & l_{22} & & 0 & \\ l_{31} & l_{32} & l_{33} & & \\ \vdots & & & \ddots & \\ l_{n1} & l_{n2} & l_{n3} & \cdots & l_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$$

$$\begin{cases} l_{11}x_1 = b_1 \\ l_{21}x_1 + l_{22}x_2 = b_2 \\ l_{31}x_1 + l_{32}x_2 + l_{33}x_3 = b_3 \\ \vdots \\ l_{n1}x_1 + l_{n2}x_2 + l_{n3}x_3 + \ldots + l_{nn}x_n = b_n \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = \dfrac{b_1}{l_{11}} \\ x_2 = \dfrac{b_2 - l_{21}x_1}{l_{22}} \\ x_3 = \dfrac{b_3 - l_{31}x_1 - l_{32}x_2}{l_{33}} \\ \vdots \\ x_n = \dfrac{b_n - l_{n1}x_1 - l_{n2}x_2 - \ldots - l_{n,n-1}x_{n-1}}{l_{nn}} \end{cases}$$

$$\boxed{i = 1:n \; : \; x_i = \frac{b_i - \sum_{k=1}^{i-1} l_{ik} \cdot x_k}{l_{ii}}}$$

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Crout - deducere formulă → $\boxed{u_{ii} = 1}$, $i = \overline{1:n}$

$A = L \cdot U$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & & \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & & & & \\ l_{21} & l_{22} & & & \\ l_{31} & l_{32} & l_{33} & & \\ \vdots & \vdots & \vdots & \ddots & \\ l_{n1} & l_{n2} & l_{n3} & \dots & l_{nn} \end{bmatrix} \cdot \begin{bmatrix} 1 & u_{12} & u_{13} & \dots & u_{1n} \\ & 1 & u_{23} & \dots & u_{2n} \\ & & 1 & \dots & u_{3n} \\ & & & \ddots & \vdots \\ & & & & 1 \end{bmatrix}$$

1 2 3 4 5 6

$a_{11} = l_{11}$

$a_{21} = l_{21}$

$a_{31} = l_{31}$

$\vdots$

$a_{n1} = l_{n1}$

→ începi pe verticală

$a_{12} = l_{11} \cdot u_{12}$

$a_{13} = l_{11} \cdot u_{13}$

$\vdots$

$a_{1n} = l_{11} \cdot u_{1n}$

$a_{22} = l_{21} \cdot u_{12} + l_{22}$

$a_{32} = l_{31} \cdot u_{12} + l_{32}$

$\vdots$

$a_{n2} = l_{n1} u_{12} + l_{n2}$

$a_{23} = l_{21} \cdot u_{13} + l_{22} \cdot u_{23}$

$a_{2n} = l_{21} \cdot u_{1n} + l_{22} \cdot u_{2n}$

→ se determină o coloană a lui L, apoi o linie a lui U și tot așa

$$j = \overline{1:n}$$

$$i = \overline{j:n}$$

$$l_{ij} = a_{ij} - \sum_{k=1}^{j-1} l_{ik} \cdot u_{kj}$$

$$i = \overline{j+1:n}$$

$$u_{ji} = \frac{a_{ji} - \sum_{k=1}^{j-1} l_{jk} \cdot u_{ki}}{l_{jj}}$$

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Doolittle - deducere formula → $l_{ii} = 1$, $i = \overline{1:n}$

$A = L \cdot U$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & & & & \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} 1 & & & & 0 \\ l_{21} & 1 & & & \\ l_{31} & l_{32} & 1 & & \\ \vdots & & & \ddots & \\ l_{n1} & l_{n2} & l_{n3} & \cdots & 1 \end{bmatrix} \cdot \begin{bmatrix} u_{11} & u_{12} & u_{13} & \cdots & u_{1n} \\ & u_{22} & u_{23} & \cdots & u_{2n} \\ & & u_{33} & \cdots & u_{4n} \\ & 0 & & \ddots & \vdots \\ & & & & u_{nn} \end{bmatrix}$$

→ începi pe orizontală

$a_{11} = u_{11}$

$a_{12} = u_{12}$

$a_{13} = u_{13}$

$\vdots$

$a_{1n} = u_{1n}$

$a_{21} = l_{21} \cdot u_{11}$

$a_{31} = l_{31} \cdot u_{11}$

$\vdots$

$a_{n1} = l_{n1} \cdot u_{11}$

$a_{22} = l_{21} \cdot u_{12} + u_{22}$

$a_{23} = l_{21} \cdot u_{13} + u_{23}$

$\vdots$

$a_{2n} = l_{21} \cdot u_{1n} + u_{2n}$

$a_{32} = l_{31} \cdot u_{12} + l_{32} \cdot u_{22}$

$\vdots$

$a_{n2} = l_{n1} \cdot u_{12} + l_{n2} \cdot u_{22}$

→ se determină o linie a lui U, apoi o coloană a lui L și tot așa

$i = \overline{1:n}$

$j = \overline{i:n}$

$$u_{ij} = a_{ij} - \sum_{k=1}^{i-1} l_{ik} \cdot u_{kj}$$

$j = \overline{i+1:n}$

$$l_{ji} = \frac{a_{ji} - \sum_{k=1}^{i-1} l_{jk} \cdot u_{ki}}{u_{ii}}$$

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Cholesky – deducere formule → $U = L^T$

$$A = L \cdot L^T$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & & & & \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & & & & 0 \\ l_{21} & l_{22} & & & \\ l_{31} & l_{32} & l_{33} & & \\ \vdots & & & \ddots & \\ l_{n1} & l_{n2} & l_{n3} & \cdots & l_{nn} \end{bmatrix} \cdot \begin{bmatrix} l_{11} & l_{21} & l_{31} & \cdots & l_{n1} \\ & l_{22} & l_{32} & \cdots & l_{n2} \\ & & l_{33} & \cdots & l_{n3} \\ & 0 & & \ddots & \vdots \\ & & & & l_{nn} \end{bmatrix}$$

$a_{11} = l_{11}^2$

$a_{12} = l_{11} \cdot l_{21}$

$a_{13} = l_{11} \cdot l_{31}$

$\vdots$

$a_{1n} = l_{11} \cdot l_{n1}$

$a_{22} = l_{21}^2 + l_{22}^2$

$a_{23} = l_{21} \cdot l_{31} + l_{22} \cdot l_{32}$

$\vdots$

$a_{2n} = l_{21} \cdot l_{n1} + l_{22} \cdot l_{n2}$

$a_{33} = l_{31}^2 + l_{32}^2 + l_{33}^2$

$a_{3n} = l_{31} \cdot l_{n1} + l_{32} \cdot l_{n2} + l_{33} \cdot l_{n3}$

$$j = 1 : n$$

$$l_{jj} = \sqrt{a_{jj} - \sum_{k=1}^{j-1} l_{jk}^2}$$

$$i = j+1 : n$$

$$l_{ij} = \frac{a_{ji} - \sum_{k=1}^{j-1} l_{jk} \cdot l_{ik}}{l_{jj}}$$

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Demonstrați proprietăți matrice ortogonală!

1) $\det Q = \pm 1$

$Q \cdot Q^T = I_4 \Rightarrow \det(Q \cdot Q^T) = 1 \Rightarrow$

$\Rightarrow \det Q \cdot \det Q^T = 1$

$\det Q = \det Q^T, \forall Q \in \mathbb{R}^{4\times 4} \Rightarrow$

$\Rightarrow (\det Q)^2 = 1 \Rightarrow \det Q = \pm 1$

5) $\|QA\|_2 = \sup_{x\neq 0} \frac{\|QAx\|_2}{\|x\|_2} \overset{2)}{=}$

$= \sup_{x\neq 0} \frac{\|Ax\|_2}{\|x\|_2} =$

$= \|A\|_2$

2) $\|Qx\|_2 = \|x\|_2$

$\|Qx\|_2 = \sqrt{\langle Qx, Qx\rangle} = \sqrt{(Qx)^T \cdot Qx} =$

$= \sqrt{x^T \cdot \underbrace{Q^T \cdot Q}_{I_n} x} = \sqrt{x^T \cdot x} =$

$= \sqrt{\langle x, x\rangle} = \|x\|_2$

3) $\|Q\|_2 = 1$

$\|Q\|_2 = \sup_{x\neq 0} \frac{\|Qx\|_2}{\|x\|_2} \overset{2)}{=} \sup_{x\neq 0} \frac{\|x\|_2}{\|x\|_2} = 1$

4) $\operatorname{cond} Q = 1$

$\operatorname{cond} Q = \|Q\|_2 \cdot \|Q^{-1}\|_2 = 1 \cdot \|Q^{-1}\|_2$

$\|Q^{-1}\|_2 = \|Q^T\|_2 = \sup_{x\neq 0} \frac{\|Q^T x\|_2}{\|x\|_2}$

$\|Q^T x\|_2 = \sqrt{\langle Q^T x, Q^T x\rangle} = \|x\|_2$ (ca mai sus)

$\Rightarrow \|Q^{-1}\|_2 = 1 \Rightarrow \operatorname{cond} Q = 1$

Gram-Schmidt-deducere formule

$u_1 = a_1$; $q_1 = \frac{u_1}{\|u_1\|_2}$

$u_2 = a_2 - \text{proj}_{u_1} a_2$; $q_2 = \frac{u_2}{\|u_2\|_2}$

$u_3 = a_3 - \text{proj}_{u_1} a_3 - \text{proj}_{u_2} a_3$; $q_3 = \frac{u_3}{\|u_3\|_2}$

$$\text{proj}_{u_j} a_i = \frac{\langle u_j, a_i \rangle}{\langle u_j, u_j \rangle} \cdot u_j = \frac{a_i^T \cdot u_j}{\|u_j\|_2^2} \cdot u_j =$$

$$= a_i^T \cdot \frac{u_j}{\|u_j\|_2^2} \cdot \frac{u_j}{\|u_j\|_2^2} = a_i^T \cdot q_j \cdot q_j =$$

$$= \langle q_j, a_i \rangle \cdot q_j$$

$$\Rightarrow u_i = a_i - \sum_{k=1}^{i-1} \langle q_k, a_i \rangle q_k \; , \; q_i = \frac{u_i}{\|u_i\|_2}$$

$(i = 1:n)$

$$\Rightarrow a_i = u_i + \sum_{k=1}^{i-1} \langle q_k, a_i \rangle q_k$$

$$q_i = \frac{u_i}{\|u_i\|_2} \Rightarrow u_i = \|u_i\|_2 \cdot q_i \Big| \Rightarrow$$

$$\Rightarrow a_i = q_i \|u_i\|_2 + \sum_{k=1}^{i-1} \langle q_k, a_i \rangle q_k \; , \; i = 1:n \quad (1)$$

$A = Q \cdot R$

$$[a_1 \; a_2 \; \dots \; a_n] = [q_1 \; q_2 \; \dots \; q_n] \cdot \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ & r_{22} & \dots & r_{2n} \\ & & \ddots & \\ & & & r_{nn} \end{bmatrix} \quad (2)$$

$(1), (2) \Rightarrow \begin{cases} r_{ii} = \|u_i\|_2 \\ r_{ij} = \langle q_i, a_j \rangle \end{cases}$

Demonstrații proprietăți H_p:

1) H_p simetrică

$$H_p^T = \left(I_m - 2\cdot\frac{v_p v_p^T}{\underbrace{v_p^T v_p}_{\|v_p\|_2^2}}\right)^T = I_m^T - 2\cdot\frac{1}{\|v_p\|_2^2}\cdot(v_p v_p^T)^T =$$

$$= I_m - \frac{2}{v_p^T v_p}\cdot(v_p^T)^T\cdot v_p^T =$$

$$= I_m - \frac{2}{v_p^T v_p}\cdot v_p\cdot v_p^T = H_p \quad \text{ectd.}$$

2) H_p ortogonală

$$H_p\cdot H_p^T \overset{1)}{=} H_p\cdot H_p = \left(I_m - \frac{2v_p v_p^T}{v_p^T v_p}\right)\left(I_m - \frac{2v_p v_p^T}{v_p^T v_p}\right) =$$

$$= I_m^2 - \frac{4v_p v_p^T}{v_p^T v_p} + \frac{4v_p(v_p^T v_p)v_p^T}{(v_p^T v_p)v_p^T v_p} =$$

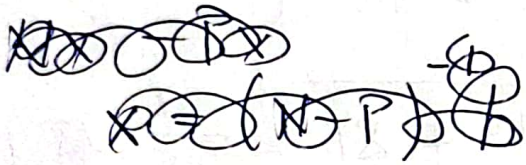
$$= I_m - \frac{4v_p v_p^T}{v_p^T v_p} + \frac{4v_p v_p^T}{v_p^T v_p} = I_m \quad \text{ectd.}$$

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Metode iterative
Sisteme de ec. liniare

$$A = N - P$$

$$(N-P)x = b$$



$$Nx - Px = b$$

$$Nx = Px + b$$

$$x^{(p+1)} = \underbrace{N^{-1}P}_{G} x^{(p)} + \underbrace{N^{-1}b}_{c}$$

$$x^{(p+1)} = Gx^{(p)} + c$$

$$\rho(G) < 1 \Rightarrow \text{converge}$$

$$A = D - L - U$$

Jacobi: $N = D$

$P = L + U$

$$\Rightarrow x^{(p+1)} = \underbrace{D^{-1}(L+U)}_{G_J} x^{(p)} + D^{-1}b$$

$$x_i^{(p+1)} = \frac{b_i - \sum_{j=1}^{i-1} a_{ij} \cdot x_j^{(p)} - \sum_{j=i+1}^{n} a_{ij} \cdot x_j^{(p)}}{a_{ii}} \qquad i = 1:n$$

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```
function [xp] = Jacobi ( A , b , tol , x0, maxiter )
    [n n] = size(A);
    x = zeros( n, 1 );
    for p = 1 : maxiter
        for i = 1 : n
            x = ( b(i) - A(i, 1 : j -1) * x0(1 : i-1) -
                A(i, i+1 : n) * x0(i+1 : n) /
                A (i,i );
        endfor
        if norm(x - x0) < tol
            return;
        endif
        x0 = x;
    endfor
endfunction
```

Gauss - Seidel :

$(N-P)x = b \Rightarrow x^{(p+1)} = N^{-1}P x^{(p)} + N^{-1}b$

$A = D - L - U$

$N = D - L$

$P = U$

$\Rightarrow x^{(p+1)} = \overbrace{(D-L)^{-1}U}^{GGS} x^{(p)} + (D-L)^{-1}b$

$$x_i^{(p+1)} = \frac{b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(p+1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(p)}}{a_{ii}}$$

$p = 1 : \text{maxiter}$

$i = 1 : n$

Suprarelaxare :

$$x_{SOR} = \omega \cdot x_{GS}^{(p+1)} + (1-\omega) \cdot x_{SOR}^{(p)}$$

$$\omega \in (0, 2)$$

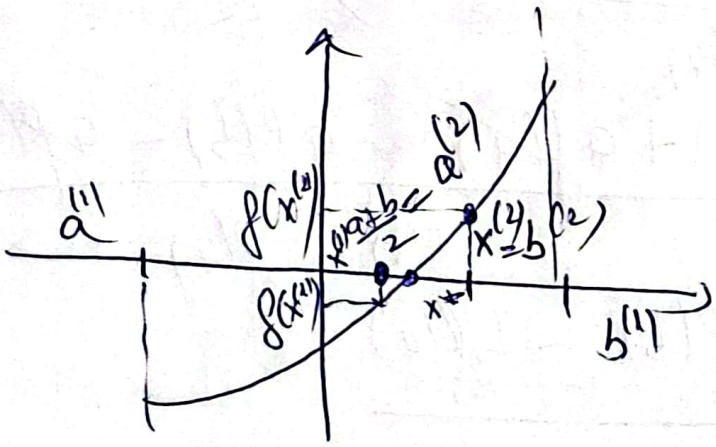
```
function [x p] = SOR(A, b, x0, tol, maxiter, w)
[n n] = size(A);
x = zeros(n, 1);
for p = 1 : maxiter
  for i = 1 : n
    x(i) = (b(i) - A(i, 1:i-1) * x(1:i-1) -
            A(i, i+1:n) * x0(i+1:n)) / A(i,i)
  endfor
  x = w*x + (1-w)*x0;
  if norm(x - x0) < tol
    return;
  endif
  x0 = x;
endfor
endfunction
```

Metoda bisectiei

$$x^{(n)} = a^{(n)} + \frac{b^{(n)} - a^{(n)}}{2}$$

function [x y] = bisectie(f, a, b, tol, maxiter)

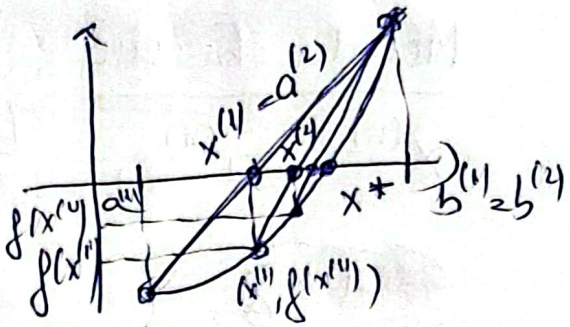
for i = 1 : maxiter
    x = a + (b - a) / 2;
    if f(x) < tol || abs(b - a) < tol
        return ;
    endif
    if f(a) * f(x) < 0
        b = x;
    else
        a = x;
    endif
endfor
endfunction



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Metoda secantei:

$$\frac{y - f(a)}{f(b) - f(a)} = \frac{x - a}{b - a}$$



$$(y - f(a))(b - a) = (x - a)(f(b) - f(a))$$

$$y(b - a) = f(a)(b - a) + (x - a)(f(b) - f(a))$$

$$y(b - a) = b \cdot f(a) - x \cdot f(b) - x \cdot f(a) - a \cdot f(b)$$

$$y = \frac{b \cdot f(a) - a \cdot f(b) - x(f(a) + f(b))}{b - a}$$

$$y - f(a) = \frac{x - a}{b - a}(f(b) - f(a))$$

$$\cap Ox : 0 = f(a) + \frac{x - a}{b - a}(f(b) - f(a))$$

$$\frac{-f(a)}{f(b) - f(a)} = \frac{x - a}{b - a}$$

$$\frac{(b - a)(-f(a))}{f(b) - f(a)} = x - a$$

$$x = \frac{-bf(a) + af(a) + af(b) - af(a)}{f(b) - f(a)}$$

$$x = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

```
function [x p] = secanta(a, b, f, tol, maxiter)
    for p = 1 : maxiter
        x = (a*f(b) - b*f(a)) / (f(b) - f(a));
        if f(x) < tol || abs(b-a) < tol
            return;
        endif
        if f(a) * f(x) < 0
            b = x;
        else
            a = x;
        endif
    endfa
endfunction
```

Metoda tangentei:

$y = f(x_0) + f'(x_0)(x - x_0)$

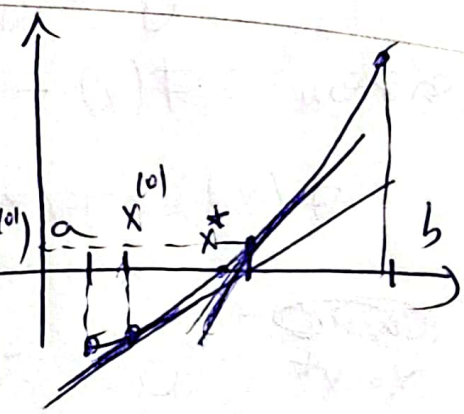
$\cap Ox: \; 0 = f(x_0) + f'(x_0)(x - x_0)$

$-f(x_0) = f'(x_0)(x - x_0)$

$x - x_0 = -\frac{f(x_0)}{f'(x_0)}$

$x = x_0 - \frac{f(x_0)}{f'(x_0)}$

$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$



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